

Paul Prae

Principal AI Engineer & Architect | Buford, GA | hireme@paulprae.com | [LinkedIn](#) | [GitHub](#) | [paulprae.com](#)

PROFESSIONAL SUMMARY

AI engineering leader with 15 years of experience delivering enterprise-scale AI, ML, and data solutions for healthcare, life science, and financial services organizations. Established track record at Amazon Web Services, Microsoft, Booz Allen Hamilton, and Slalom Consulting — architecting production AI systems for Fortune 500 clients across HIPAA and FedRAMP-compliant environments. Currently builds AI agents and manages petabyte-scale healthcare data infrastructure at Arine, a WEF Technology Pioneer covering 50M+ health plan members. Combines full-stack technical depth — spanning data engineering, ML pipelines, and cloud architecture — with progressive engineering leadership, including training teams of up to 100 engineers on AI-first development and growing a consulting practice to \$1.4M ARR.

PROFESSIONAL EXPERIENCE

Staff AI DataOps Engineer

Arine | Remote | Sep 2025 – Present

- Designed the AI DataOps Toolkit — an agentic development framework with AI governance standards, reference architectures, and change management procedures — for a WEF Technology Pioneer whose medication-optimization platform serves 45+ health plans covering 50M+ members.
- Developed HIPAA-compliant coding assistants including a Data Engineering Agent that autonomously generates, deploys, and tests ETL pipelines and a Dev Environment Setup Agent that eliminated multi-day onboarding processes.
- Led AI enablement workshops training groups of up to 100 engineers on AI-first development and agentic coding workflows using Cursor AI, Claude Code, and GitHub Copilot.
- Built an observability pipeline for Arine's autonomous pharmacist AI agent — ingesting production behavior data from DynamoDB into Snowflake, transforming with dbt, and surfacing performance metrics in QuickSight dashboards.
- Managed the enterprise data platform on Snowflake and AWS, processing petabytes of healthcare data from hundreds of heterogeneous sources supporting downstream ML and AI systems core to the company's products.

Chief AI Architect, Senior Manager

Booz Allen Hamilton | Atlanta, GA | Jul 2024 – Mar 2025

- Architected AI and analytics solutions at a Fortune 500 consulting firm (\$9B+ revenue), enabling healthcare and life science clients to harness clinical, behavioral, and genomic data for population health and predictive medicine.
- Designed and deployed AI software products across FedRAMP and HIPAA-compliant environments, collaborating with cross-functional teams of engineers, data scientists, and domain experts.
- Mitigated algorithmic bias and data privacy risks in regulated healthcare AI deployments, ensuring compliance with clinical and regulatory requirements.
- Advised healthcare and life science leadership on AI transformation strategy, translating emerging technology capabilities into actionable roadmaps for clinical decision support.

Neuroinformatics Data Platform Architect

TRENDS Center | Atlanta, GA | Jan 2022 – Sep 2023

- Led architecture for COINSTAC, an open-source federated analysis platform for neuroimaging and genomics data with differential privacy, resulting in a peer-reviewed publication in *Frontiers in Neuroinformatics*.
- Architected AWS cloud infrastructure including GPU-focused compute for deep learning; served as Scrum master driving Agile ceremonies and cross-team collaboration.
- Drove platform commercialization through user adoption efforts, open-source contributor recruitment, SBIR grant writing, and industry partnerships.

Chief AI Officer, Founder

Hyperbloom | United States | Jan 2020 – Sep 2025

- Founded and scaled an AI consulting practice to \$1.4M ARR from a \$40K investment, delivering cloud-based AI and data solutions to healthcare, life science, and financial services clients.
- Architected disaster recovery and data governance for clinical trial software supporting 10,000+ research sites across 45 countries with 5.5M monthly remote monitoring activities.
- Led teams of solutions architects and account managers through full sales cycles; trained engineers on entrepreneurship and built decentralized talent platforms for independent contractors.

Enterprise AI and ML Solutions Architect

Amazon Web Services | Atlanta, GA | Aug 2018 – May 2021

- Managed a portfolio of 10+ enterprise accounts — including Cox Enterprises, Equifax, NCR, Accenture, and Deloitte — designing and integrating AI/ML solutions on AWS using SageMaker, Lex, and Comprehend.
- Engaged nationally as an AI/ML subject matter expert, authoring technical white papers and reference architectures and delivering production-ready proof-of-concept solutions.
- Partnered with C-suite executives to define technical and business strategies spanning go-to-market planning, cost optimization, and technical recruiting.

Senior AI Engineer

NeuroLex Labs | Atlanta, GA | Jan 2018 – May 2020

- Architected automated ML pipelines for voice-based health diagnostics, processing text and audio data to support prediction and prevention of neurodegenerative diseases.
- Mentored international research fellows in the Tribe program and drove test-driven development practices across the engineering team.

Advanced Analytics Consultant

Slalom Consulting | Atlanta, GA | Jul 2015 – Jan 2018

- Delivered 7+ analytics engagements including a behavioral healthcare application generating \$2M+ in revenue and statewide demand forecasting of crisis services across dozens of Georgia facilities.
- Recognized nationally as a subject matter expert in NLP and ML; delivered public presentations on Azure Machine Learning, conversational AI, and cloud bot architecture.
- Operationalized data science teams by transitioning experiments from isolated environments to automated, cloud-based production systems supporting seamless collaboration.

Software Engineer

Red Ventures | Fort Mill, SC | Mar 2014 – Jul 2015

- Engineered full-stack marketing platform serving 16+ Fortune 500 clients including DirecTV, Verizon, AT&T, and Liberty Mutual across hundreds of high-traffic consumer-facing websites.
- Won first place at HACKATHON clt, processing 140M+ rows of purchase data using Apache Spark and map-reduce algorithms for a crowdsourced delivery system.

Senior Support Engineer

Microsoft | Charlotte, NC | Jul 2012 – Mar 2014

- Delivered advanced troubleshooting for hundreds of high-impact web applications and server farms serving tens of thousands of users at Fortune 100 clients including Chevron, Walmart, Disney, and BlueCross BlueShield.
- Nominated for Global Business Support’s “Entrepreneur of the Year” (2013); graduated from the Microsoft Academy for College Hires (MACH) program.

EDUCATION

BS, Computer Science | BA, Cognitive Science

University of Georgia | 2008 – 2012 Both degrees with specialization in Artificial Intelligence. Graduated Cum Laude. Dean’s List.

TECHNICAL SKILLS

AI & Machine Learning: Generative AI, Large Language Models (LLMs), Multi-Agent Systems, Natural Language Processing, Deep Learning, MLOps, Predictive Analytics, Conversational AI, Prompt Engineering

Cloud & Infrastructure: AWS (Bedrock, SageMaker, ECS/EKS, Lambda, Step Functions, S3, Glue, DynamoDB, QuickSight), Microsoft Azure, Snowflake, Docker, Distributed Systems

Data Engineering: ETL/CDC Pipelines, dbt, Apache Kafka, PostgreSQL, Neo4j, Data Governance, Data Privacy (HIPAA, GDPR, CCPA)

Programming Languages: Python, SQL, JavaScript, TypeScript, C#, Bash, Rust

AI Tools & Frameworks: Cursor AI, Claude Code, GitHub Copilot, LangChain, Ollama, n8n, Open WebUI, MCP Tools

Leadership & Strategy: Engineering Team Management, Solutions Architecture, AI Governance, Executive Coaching, Agile/Scrum, Technical Recruiting

CERTIFICATIONS

- **AWS Certified Machine Learning Engineer** — Amazon Web Services (Sep 2025)
- **AWS Certified AI Practitioner** — Amazon Web Services (Sep 2025)

- **AWS Certified Machine Learning – Specialty** – Amazon Web Services (Jan 2020)
- **AWS Certified Solutions Architect** – Amazon Web Services (Jul 2018)

PROJECTS

[Modular Earth](#)

Not-for-profit organization developing open-source AI agents to help working-class entrepreneurs generate wealth.

Architectures enforce local-first, privacy-preserving, and decentralized design principles using Ollama, LangChain, n8n, and Neo4j. Vision: deploy AI-as-a-public-service in libraries nationwide at zero cost to job seekers.

[Neona](#)

Educational conversational AI agent that teaches artificial intelligence concepts through interactive dialogue. Demonstrates end-to-end chatbot architecture from natural language understanding design to cloud deployment.

PUBLICATIONS

[Enhancing Collaborative Neuroimaging Research: Introducing COINSTAC Vaults for Federated Analysis and Reproducibility](#)

Frontiers in Neuroinformatics (Jun 2023). Co-authored peer-reviewed paper presenting COINSTAC Vaults for federated neuroimaging analysis, enabling researchers to analyze datasets without sharing raw data while improving reproducibility and increasing sample sizes across research institutions.